

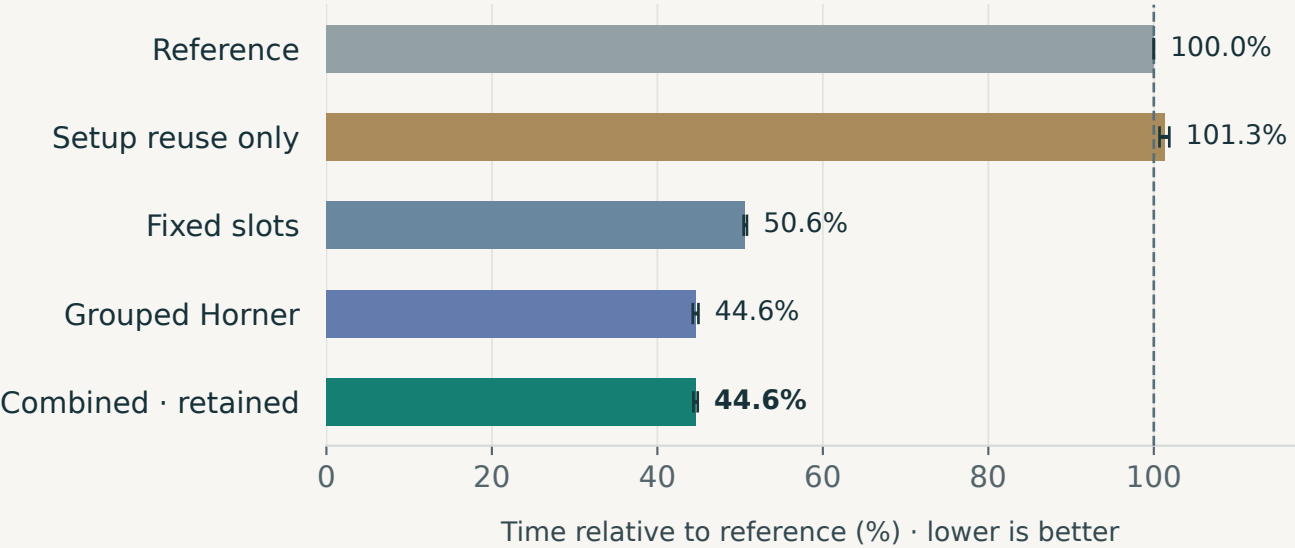
Where the performance gains actually are

ECC index calculus / Summation polynomials / Frozen measurements · 14 September 2026

55.4% less specialization time

A · Per-target specialization

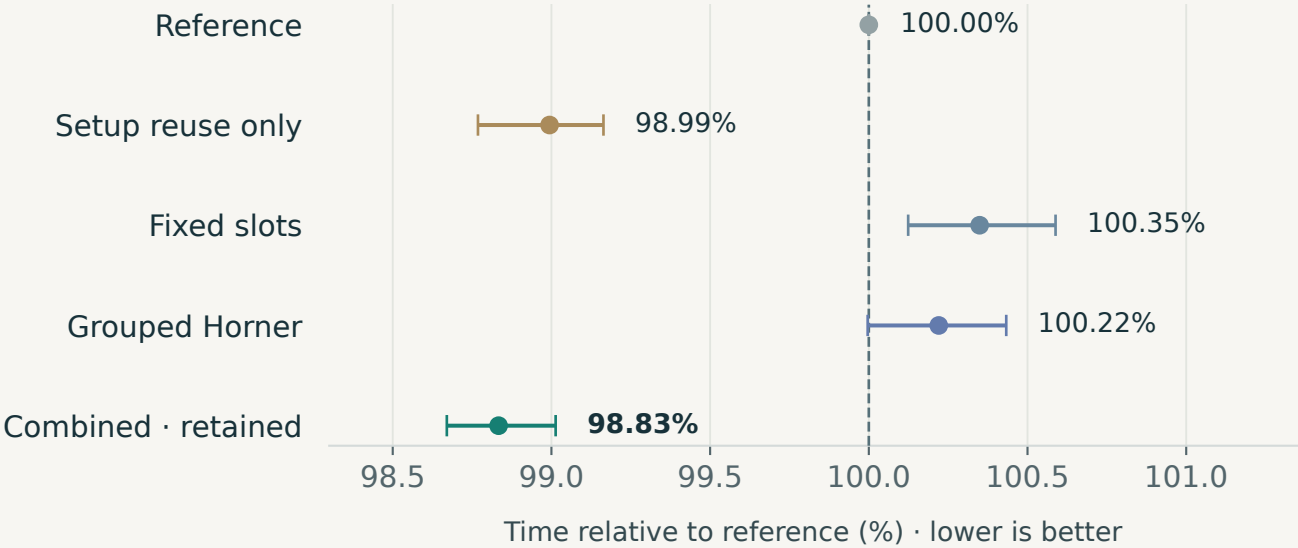
Reference = 100%; whiskers show paired 95% intervals



Full-DLP improvement remains unproven

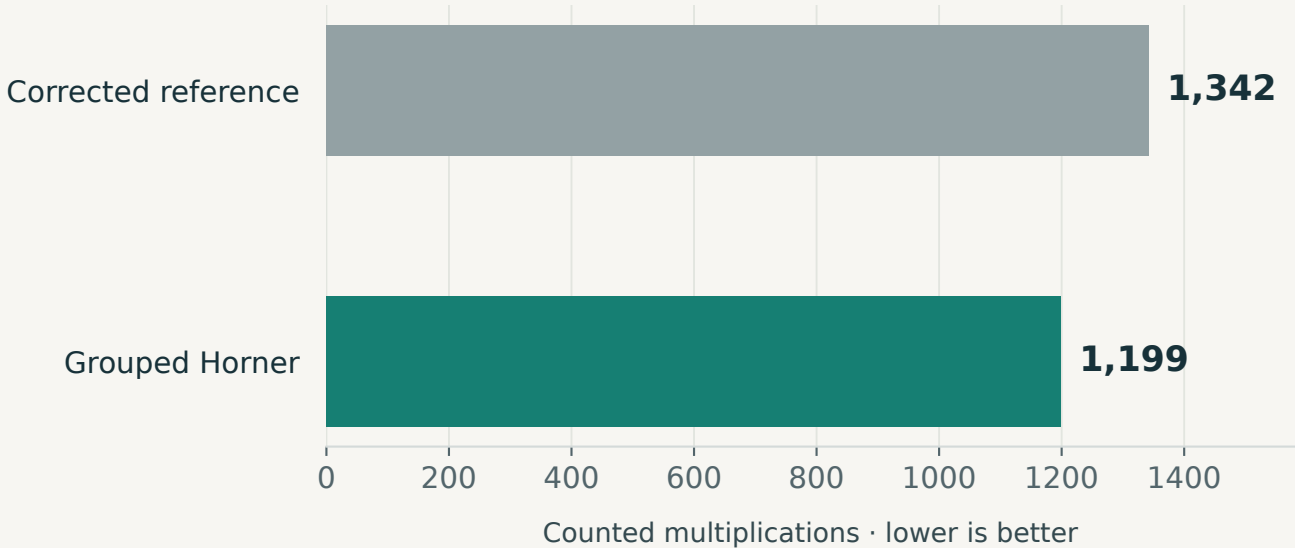
B · Once-per-curve symbolic setup

Zoomed scale highlights the small setup change



C · Real arithmetic savings

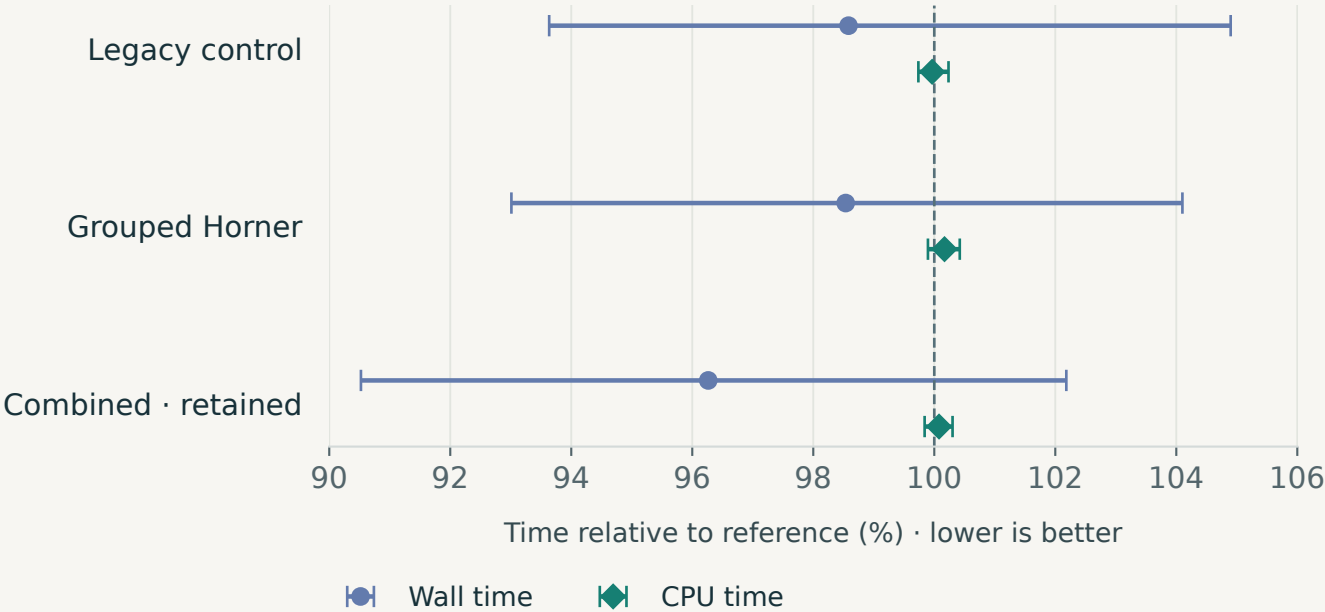
Fp multiplications per specialization · p = 67, seed = 11



143 fewer products (10.7%). Historical 1,770 estimate excluded: its correction was accounting, not an optimization gain.

D · Full-DLP confirmation

All intervals cross 100%: no established gain or regression



Evidence: 342 ablation runs + 140 confirmation runs. Polynomial and scalar checks passed throughout.

Paired 95% intervals cluster repetitions by input. Shared host; small corpus. Local gains are not additive or an asymptotic advance.